

Ryan Zomorodi

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EDUCATION

University of Illinois Chicago

Master of Science, Biostatistics

- Cumulative GPA: 4.0/4.0

Chicago, IL

Aug 2023 — May 2025

University of Illinois Chicago

Bachelor of Science, Biological Science

- Minors: Computer Science & Public Health
- Graduated with Highest Distinction
- Honors College
- Cumulative GPA: 4.0/4.0

Chicago, IL

Aug 2019 — May 2023

WORK EXPERIENCE

Children's Environmental Health Initiative

Research Specialist (Full Time)

Chicago, IL

Jul 2025 — Present

- Led data acquisition, processing, and validation efforts for large nationwide health and environmental exposure datasets (exceeding 1TB) using R, Python, and SQL.
- Utilized longitudinal modelling to analyze Epic electronic health records (EHR) data to explore the associations between air pollution and cardiovascular and respiratory health outcomes.
- Built an ArcGIS Pro Python toolbox to simplify the process of generating reliable, geospatially resolved Bayesian estimates of chronic disease, now utilized by state health departments. Presented work at Council of State and Tribal Epidemiologists.
- Developed a set of internal dashboards to investigate excess mortality and morbidity among the elderly following extreme weather events using DuckDB and ObservableJS.
- Trained over 200 state, tribal, and local public health professionals in collaboration with CDC on the use of geospatial analytics to assess and address chronic health disparities.
- Designed and maintained the lab's online geospatial analytics training program materials and website.

Research Associate (Full Time)

Jun 2023 — Jul 2025

- Identified opportunities for automation within existing data workflows and built four internal R packages streamlining data acquisition and analysis processes for myself and across the broader team.
- Automated the collection of datasets through the use of web scraping, API requests, and SQL queries to support epidemiological research.
- Integrated precipitation, wind, and power outage datasets to develop estimates of climate disaster exposure and evaluate associated health risks.
- Performed analysis of hospital and physician accessibility utilizing two-step floating catchment area methods.

Department of Rheumatology, University of Illinois Chicago

Research Assistant (Part Time)

Chicago, IL

Jul 2018 — May 2023

- Independently performed bioinformatic analysis of RNA-seq data including differential expression, weighted network gene correlation analysis (WGNCA), Passing Attributes between Networks for Data Assimilation (PANDA) using R and Python.
- Conducted and oversaw experimental techniques such as ELISA and qRT-PCR.

Walgreens

Pharmacy Technician (Part Time)

Arlington Heights, IL

July 2021 — May 2023

- Utilized electronic health records software to access patient medical records, input insurance, allergy, and health condition records, and resolve patient inquiries.
- Resolved insurance inquiries and resubmitted claims in order to ensure patient prescription adherence.
- Filled, mixed, and dispensed prescriptions for approval by pharmacists.

PROJECTS

GeoAdvocates

Chicago, IL

Co-Lead

Dec 2021 — Jun 2024

- Designed, in collaboration with another student, a two-part workshop series curriculum to teach students within multiple classes outlining medical deserts within Chicago and how students can utilize Geographic Information Systems (GIS) to expose Chicago health disparities.
- Presented to several UIC courses (SOC251, PUBH100, US101, SOC100, and PPOL100) and registered student organizations.
- Designed, collected, and analyzed survey data taken from participants before and after each workshop.
- Discussed findings at conferences including the UIC Minority Health Conference and UIC School of Public Health Research and Scholarship Week Lightning Talks.

Scientific Software

- **Rate Stabilizing Toolbox (RSTbx)** - <https://cehicapacitybuilding.org/RateStabilizingToolbox/>
Python toolbox for generating reliable, local-level age-standardized measures of chronic disease using the restricted univariate conditional autoregressive (UCAR) bayesian model.
- **RSTr** - <https://cehicapacitybuilding.org/RSTr/>
R package for generating reliable, local-level age-standardized measures of chronic disease using several different conditional autoregressive bayesian models. This package allows for greater insight into the modeling process than the RSTbx.
- **exposhur** - <https://cehicapacitybuilding.org/exposhur/>
Internal R package for the Children's Environmental Health Initiative which allows for the calculation of hurricane exposure measures at highly resolved geospatial scales.

Open Source Software

I use open-source software every day, and I make it a point to give back by maintaining and contributing to open-source projects whenever I can.

- **sasquatch** - <https://docs.ropensci.org/sasquatch/>
R package that bridges SAS and R. Utilizes the Python package **SASPy** to enable transfer of data between R and SAS, interactive SAS evaluation, basic remote SAS client file management, and rendering of SAS within quarto/rmarkdown documents.
- **legs** - <https://ryanzomorrodi.github.io/legs>
Terminal User Interface (TUI) for viewing R objects like `data.frames`, `matrixs`, `arrays`, `lists`, and `vectors`. Designed to allow quick exploration of deeply nested R objects within a neovim R session.
- **arcpygris** - <https://github.com/ryanzomorrodi/arcpygris>
Python toolbox for downloading Census boundaries into an ArcGIS Pro project. Inspired heavily by the R package **tigris**.
- **socratadata** - <https://ryanzomorrodi.github.io/socratadata>
R package API wrapper that utilizes Rust to parse Socrata datasets into R dataframes. Also features support for Socrata Query Language (SoQL) and the Socrata Discovery API.
- **centr** - <https://ryanzomorrodi.github.io/centr>
R package designed to calculate planar or geodesic weighted mean or median spatial centers.
- **healthatlas** - <https://ryanzomorrodi.github.io/healthatlas>
R package API wrapper for Metopio health atlas data. Allows for easy exploration of health atlas datasets and metadata.

PUBLICATIONS

Variation in One-Year Mortality Following Severe Weather Exposure Among Older Americans by Chronic Health Condition and Sociodemographic Status

Journal of the American Geriatrics Society

Feb 2026

[10.1111/jgs.70237](https://doi.org/10.1111/jgs.70237)

Bayesian Models to Generate Small Area Estimates of Population Health: Tutorial for Using Rate Stabilizing Tools and Their Output

Journal of Medical Internet Research Public Health Surveillance

Jan 2026

[10.2196/83498](https://doi.org/10.2196/83498)

Significance of IL-34 and SDC-1 in the pathogenesis of RA cells and preclinical models

Clinical Immunology

May 2023

[10.1016/j.clim.2023.109635](https://doi.org/10.1016/j.clim.2023.109635)

Inhibition of IRAK4 dysregulates SARS-CoV-2 spike protein-induced macrophage inflammatory and glycolytic reprogramming

Cellular and Molecular Life Sciences

May 2022

[10.1007/s00018-022-04329-8](https://doi.org/10.1007/s00018-022-04329-8)

Metabolic reprogramming of macrophages instigates CCL21-induced arthritis

Immunology & Cell Biology

Dec 2021

[10.1111/imcb.12512](https://doi.org/10.1111/imcb.12512)

Metabolic regulation of RA macrophages is distinct from RA fibroblasts and blockade of glycolysis alleviates inflammatory phenotype in both cell types

Cellular and Molecular Life Sciences

Oct 2021

[10.1007/s00018-021-03978-5](https://doi.org/10.1007/s00018-021-03978-5)

Interleukin-34 Reprograms Glycolytic and Osteoclastic Rheumatoid Arthritis Macrophages via Syndecan 1 and Macrophage Colony-Stimulating Factor Receptor

Arthritis & Rheumatology

Sep 2021

[10.1002/art.41792](https://doi.org/10.1002/art.41792)

Tofacitinib therapy intercepts macrophage metabolic reprogramming instigated by SARS-CoV-2 Spike protein

European Journal of Immunology

Jun 2021

[10.1002/eji.202049159](https://doi.org/10.1002/eji.202049159)

CCL25 and CCR9 is a unique pathway that potentiates pannus formation by remodeling RA macrophages into mature osteoclasts

European Journal of Immunology

Jan 2021

[10.1002/eji.202048681](https://doi.org/10.1002/eji.202048681)

IRAK4 inhibition: a promising strategy for treating RA joint inflammation and bone erosion

Cellular & Molecular Immunology

May 2020

[10.1038/s41423-020-0433-8](https://doi.org/10.1038/s41423-020-0433-8)

Macrophages are the primary effector cells in IL-7-induced arthritis

Cellular & Molecular Immunology

Jun 2019

[10.1038/s41423-019-0235-z](https://doi.org/10.1038/s41423-019-0235-z)

Impact of obesity on autoimmune arthritis and its cardiovascular complications

Autoimmunity Reviews

Jun 2018

[10.1016/j.autrev.2018.02.007](https://doi.org/10.1016/j.autrev.2018.02.007)

PRESENTATIONS / WORKSHOPS

Learning to use the RSTbx to calculate high quality, local level measures of health status

NACDD Surveillance, Epidemiology, and Evaluation Learning (SEEL)

12 May 2026

Working with the Tidycensus Package Mini-Course

GIS Capacity Building Project

[Link](#)
9 — 11 Dec 2025

Beyond the eye of the storm: why localized, multi-dimensional aspects of severe weather exposure matter for vulnerable, older adult populations

CAFE: Hurricanes & Health: Understanding Risks and Fostering Resilience

4 Aug 2025

Enhancing Local Level Surveillance and Data Visualization of Health Outcomes: Introducing a Set of Rate Stabilizing Tools

Council of State and Territorial Epidemiologists (CSTE)

10 Jun 2025

Enhancing Data to Action: Approaches for Using Maps to Address Chronic Disease <i>Council of State and Territorial Epidemiologists (CSTE)</i>	9 Jun 2025
Encounter with sasquatch: Using SAS, R, and Quarto together <i>SAS Open Source Community of Practice</i>	Link 25 May 2025
Using GIS to Study Social Determinants of Health with R <i>Northern Arizona University Center for Health Equity Research</i>	Link 6 — 7 Jun 2024
An Introduction to ArcGIS Online Mini-Course <i>GIS Capacity Building Project</i>	17 — 24 Oct 2023
A Gentle Introduction to GIS Mini-Course <i>GIS Capacity Building Project</i>	19 — 26 Sep 2023
RNA-sequencing analysis of Macrophage Hyperactivation <i>UIC Undergraduate Research Forum</i>	18 Apr 2023
Empowering Community Members to Investigate and Address Chicago's Health Disparities <i>American Public Health Association</i>	9 Nov 2022
Empowering Community Members to Investigate and Address Chicago's Health Disparities <i>UIC Minority Health Conference</i>	Link 8 Apr 2022
Developing Student Workshops to Increase Community Engagement in Addressing Chicago's Health Disparities <i>UIC School of Public Health Research and Scholarship Week Lightning Talks</i>	22 Apr 2022

TEACHING EXPERIENCE

2026 GIS Training for Surveillance of Heart Disease, Stroke, and Other Chronic Diseases in State Health Departments <i>GIS Capacity Building Project</i>	18 Feb 2026 — 4 Jun 2026
2025 GIS Training for Surveillance of Heart Disease, Stroke, and Other Chronic Diseases in State Health Departments <i>GIS Capacity Building Project</i>	25 Mar 2025 — 31 Jul 2025
Hawaii State Department of Health & University of Hawai'i GIS Capacity Building <i>GIS Capacity Building Project</i>	28 Jan 2025 — 12 Jun 2025
Advanced GIS Training for Surveillance of Heart Disease, Stroke and Other Chronic Diseases Among Health Department Staff <i>GIS Capacity Building Project</i>	13 Nov 2024 - 29 Jan 2025
2024 GIS Training for Surveillance of Heart Disease, Stroke, and Other Chronic Diseases in State Health Departments <i>GIS Capacity Building Project</i>	6 Mar 2024 — 31 Jul 2024
GIS and Chronic Disease Training for Tribal Members and TEC Staff Fall 2023 <i>GIS Capacity Building Project</i>	10 Oct 2023 — 31 Jan 2024

HONORS

Highest Distinction <i>UIC Department of Biological Sciences</i>	2023
Pasteur Prize Nominee <i>UIC Department of Biological Sciences</i>	2023
Research Grant <i>UIC Honors College</i>	2022
Travel Grant <i>UIC Honors College</i>	2022

Maurice Prize <i>UIC Honors College</i>	2022
Maurice Prize <i>UIC Student Leadership and Civic Engagement</i>	2022
Undergraduate Most Outstanding Presentation <i>UIC School of Public Health Research Week</i>	2022
Chancellor’s Student Service Award <i>UIC Student Leadership and Civic Engagement</i>	2022
President’s Volunteer Service Award - Bronze <i>UIC Student Leadership and Civic Engagement</i>	2021
1st place - Division 4 <i>Collegiate Chess League</i>	2021
Chancellor’s Student Service Award <i>UIC Student Leadership and Civic Engagement</i>	2021
Research Grant <i>UIC Honors College</i>	2022
PROFESSIONAL MEMBERSHIP	
Harvard Innovation Lab	2022 — 2024
Clinton Global Initiative University Fellowship	2023